National University of Computer and Emerging Sciences

Lahore Campus

Artificial Intelligence Lab (AL-2002)

Sessional Exam

Date: April 16th 2025

Total Time (Hrs.): 1.5

Total Marks: 25

Total Questions: 2

Course Instructor(s)

Mr. Muhammad Hasan

Ms. Rida Amir

_____ Roll No

BDS – 6B, BCS – 6H Sections

_____ Student Signature

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Instructions

- Attempt all questions.
- You must ensure your submission within your total time of 1.5 Hrs.
- Submit a single file in the designated folder of your section.
- The submitted file should be named with your roll number e.g. 25L-1234
- Verify your submitted file is visible in the submission folder, and check that the file size is shown next to it (to ensure it's not an empty i.e. 0 KB file).
- Path for submission will be provided at the time of submission.

1. Genetic Algorithm – Controlled Mutation (15 Marks)

Function:

$$f(x) = -x^2 + 10x + 5$$

Tasks:

- Implement GA:
  - Population size = 10
  - Number of generations = 10
  - Mutation:

    Mutation formula:

    $$x = x + random.uniform(-1, 1)$$

    Apply mutation with:

    ```
    mutation_rate = 0.2    # 20% probability
    ```
- Selection using Ranking Method
  - Rank individuals based on fitness (highest → rank 1)
  - Select top 50% individuals based on ranking
- Run the algorithm

  For each generation, display:
  - Best individual
  - Fitness value
- Final Output

  Display:
  - Best solution after 10 generations
  - Maximum fitness achieved

- Modify:
  Now replace Ranking Selection with Roulette Wheel Selection (Fitness-Proportionate Selection)
  - Selection probability proportional to fitness
- Compare:
  Compare Ranking Selection vs Roulette Selection based on:
  - Convergence speed (i.e., how quickly the algorithm reaches near-optimal solution)
  - Stability of improvement across generations
- Explain (in comments):
  - Which selection method converged faster?
  - Which method showed more stable behavior?
  - Why does ranking help in avoiding premature convergence?

2. *Greedy vs A* Search (10 Marks)

```
graph = {
    'A': [('B',1), ('C',4)],
    'B': [('D',2), ('E',5)],
    'C': [('F',1)],
    'D': [],
    'E': [],
    'F': []
}

heuristic = {
    'A': 7,
    'B': 6,
    'C': 2,
    'D': 1,
    'E': 3,
    'F': 0
}
```

Tasks:

- Implement:
  - Greedy Best-First Search
  - A* Search
- Display:
  - Path from A → F
  - Total cost of path
  - Nodes explored
- Explain (in comments):
  - Why results differ?